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IARCO 2026 RESEARCH PROPOSAL

Microglia–T Cell Intercellular Communication as a Driver of Extracellular-Vesicle-Mediated Tau Propagation in Alzheimer's Disease Models

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Research Topic: Microglia-T Cell Intercellular Communication as a Driver of Extracellular-Vesicle-Mediated Tau Propagation and Prion-Like in Alzheimer's Models

Research Problem

Alzheimer's disease (AD) is characterized by the progressive accumulation and anatomically patterned spread of misfolded tau protein through the brain. This pattern of tau propagation looks similar to how misfolded prion proteins spread, so tau spread is often called “prion-like” [1].

Extracellular vesicles (EVs), small vesicles released by brain cells, are one experimentally supported way through which tau can move between cells. Microglia take up tau and release it again inside EVs, and blocking EV release slows tau spread in mice [2]. A recent imaging study directly showed tau filaments can be enclosed within brain EVs and physically associated with their limiting membranes, providing direct evidence that EVs can carry assembled tau species [3]. T cells have also been implicated in AD-related neuroinflammation. In a mouse model of tauopathy, T-cell infiltration promoted neuroinflammation and cognitive decline [4]. In humans, clonally expanded CD8+ T-cell populations have been identified in the cerebrospinal fluid of individuals with AD [5]. Newer studies have further implicated γδ T-cell–microglia crosstalk in AD models. In particular, γδ T-cell–microglia crosstalk has been associated with subsequent CD8+ T-cell infiltration and AD-like pathology in an amyloid mouse model [9].

These two research areas, extracellular vesicle-mediated tau spread and T cell involvement in AD, have mostly been studied separately. No study has yet asked whether the contact between glial cells and T cells itself changes how glial cells release tau-carrying EVs. This is the gap this study addresses.

Existing Literature

Vingtdeux et al. first proposed that exosomes, a specific type of EV, carry misfolded proteins between brain cells in AD, extending the prion concept to tau and amyloid-beta [1]. Asai et al. confirmed this experimentally: microglia phagocytose tau and re-release it in exosomes, and depleting microglia or blocking exosome formation slows tau spread in a mouse model of tauopathy [2]. Fowler et al. used cryo-electron tomography to show that tau filaments in AD brain tissue are physically tethered inside the membranes of brain EVs, giving direct structural proof of this pathway [3]. Astrocytes add a second route: the P301L tau mutation changes how astrocytes build EVs, and astrocyte-derived EVs carrying inflammatory cargo are linked to tau pathology and worse cognition in patients [10]. Blocking Tsg101, a protein required for EV formation, specifically in microglia slows tau spread in mice, pointing to a druggable step in this pathway [11]. Wu and Eisel reviewed how microglia and astrocytes signal to each other during AD, describing this crosstalk as central to how neuroinflammation develops [12].

A separate body of work shows that AD is not only driven by microglia and astrocytes but also by the adaptive immune system. Laurent et al. found that T cells entering the hippocampus worsen neuroinflammation and cognitive decline in a tauopathy mouse model [4]. Gate et al. found unusual, expanded populations of CD8+ T cells in the cerebrospinal fluid of AD patients, and this expansion was linked to worse cognition [5]. Unger et al. showed CD8+ T cells change neuronal and synaptic gene expression in an amyloid mouse model [6], and Altendorfer et al. showed brain CD8+ T cells take on a long-term “memory” identity with age and AD pathology [7]. Piehl et al. profiled cerebrospinal fluid immune cells across ages and found signatures linking microglia activation to T cell responses [8]. Most recently, Rezende et al. showed that

a specific subset of T cells, gamma-delta T cells, communicate directly with microglia and control how many CD8+ T cells later enter the brain in an amyloid mouse model [9].

Immunology research outside the brain gives a reason to expect that glia-T cell contact could regulate EV release specifically. When a T cell touches another immune cell, it forms a tight contact point called the immune synapse, and this contact point is a known site of directed EV release, separate from a cell's normal, constant EV shedding [14]. Cespedes et al. showed this contact-triggered EV release carries a distinct set of molecules compared to background EV release [13], and Gutierrez-Vazquez et al. showed EV exchanged at immune cell contact points can transfer proteins and RNA between the two cells [15].

Taken together, the literature establishes that glial cells release tau-carrying EVs, and that T cells enter the AD brain and communicate with glial cells. What is missing is a direct test of whether that glia-T cell contact changes the release or tau-seeding ability of glial EVs, which is the focus of this proposal.

Research Question

Does direct contact with T cells change the amount or tau-seeding activity of extracellular vesicles (EVs) released by tau-exposed microglia?

Research Topic Justification

This study is novel because it connects two research areas that are usually studied on their own: EV-mediated tau spread and T cell involvement in AD. Instead of assuming these are two separate problems, this study tests whether they are mechanistically linked through direct cell contact. If glia-T cell contact does regulate EV-associated tau release, it would point to a more specific and testable drug target than broadly blocking either T cells or EVs on their own.

Methodology

Study Design

This will be a controlled in vitro study comparing EV release from tau-exposed microglia cultured with and without direct T cell contact. A transwell insert will be used to separate glia and T cells physically while still allowing shared culture medium, so that direct-contact effects can be distinguished from effects caused only by released signaling molecules.

Cell Preparation

A mouse microglial cell line (BV2) will be used for feasibility, following the EV-release protocol established by Asai et al. [2]. Microglia will be pre-exposed to recombinant human tau protein for 24 hours to load them with tau, mirroring published tau-uptake protocols [2]. CD8+ T cells will be isolated from mouse spleen using magnetic bead sorting and activated with standard CD3/CD28 stimulation before co-culture.

Experimental Groups

Three conditions will be compared: (1) tau-loaded microglia alone, (2) tau-loaded microglia in direct contact with activated CD8+ T cells, and (3) tau-loaded microglia separated from CD8+ T cells by a transwell insert. Each condition will be run in triplicate across three independent experiments.

Extracellular Vesicle (EV) Isolation and Characterization

EVs will be isolated from conditioned media by differential ultracentrifugation, following standard protocols used in prior studies of tau-carrying EVs [2], [10]. This method isolates a mixed population of EVs rather than a confirmed, pure exosome fraction, so vesicles will be referred to as EVs throughout, in line with current reporting standards. EV size and concentration will be measured by nanoparticle tracking analysis. EV identity will be confirmed by western blot for the markers CD63 and Tsg101, and tau content will be measured by western blot using a total-tau antibody.

Tau-Seeding Assay

The ability of isolated EVs to trigger tau misfolding will be tested using a tau-biosensor cell line that produces a measurable signal when it takes up seed-competent tau, a standard assay for testing prion-like seeding activity [1]. Seeding activity will be compared across the three experimental conditions.

Statistical Analysis

EV concentration, tau content, and seeding activity will be compared across the three conditions using one-way ANOVA with post-hoc pairwise comparisons. A p-value below 0.05 will be considered statistically significant. Analysis will be performed using GraphPad Prism or R.

Ethical Considerations

The BV2 microglial cell line will be obtained from a verified repository to ensure a documented, traceable source and on arrival, cell identity will be confirmed by checking microglial morphology and marker expression, and cultures will be screened for mycoplasma

contamination by PCR before use and at regular intervals. Only mycoplasma-negative, low-passage cultures will be used, with cells replaced from frozen stock past a set passage limit.

Because this component uses an established cell line, it does not require separate ethical approval for the cell line itself. Mouse spleen tissue for T cell isolation will be obtained through institutional animal-use channels under existing ethical approval where the study is carried out, following standard humane-endpoint protocols. The study does not involve human participants.

Expected Outcomes

It is expected that microglia in direct contact with T cells will release more EVs, with higher tau content and stronger seeding activity, than microglia cultured alone or separated by a transwell. This would show that T cell contact is not just a bystander event in AD but an active driver of EV-mediated tau spread. If no difference is found between the direct-contact and transwell groups, this would instead suggest that glial EV release and T cell infiltration act independently, which is also a useful and publishable finding.

Limitations

A cell line model cannot fully represent the complex, aged brain environment where AD develops. Findings from this in vitro study would need to be confirmed in animal models before any conclusions about disease progression can be drawn. Because this design only compares contact versus no-contact, it can show a correlation between T cell contact and EV changes, but a stronger, gene-based test would be needed to prove the contact directly causes the change.

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